

Bring a dead Hard Drive back to life

Recover data from a dead harddrive by physically changing platters and saving thousands in per GB professional recovery charges

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Your hard drive though seemingly tough is in fact pretty delicate and contains precious data. HDDs have plenty of mechanical parts and the more there are, the more things can go wrong. One fine day you'll come across this...

Operating system not found

The most probable cause for this is a hard drive failure. During the numerous restarts that you'll perform after this happens, there is a likelihood that your drive may be detected again. Immediately take a backup onto any other drive you may have connected to your PC. However, when all else fails, and you desperately need your data you can swap the platters of your current drive onto a hard-drive with similar make and hope to recover it.

Before you go about doing so, you'll have to figure out where exactly the problem lies. If it so happens that your logic board on your hard drive is wrecked then you'll have to find a similar logic board and transfer it onto the drive. Depending on the severity of your

backup all your data. There are a few criteria that need to be satisfied in order to get a matching PCB. Check box. You can buy a spoilt HDD and use its PCB. If you can't find the needed PCB easily then there are specialty stores that stock just PCBs and with any luck you'll find the one you need, for about ₹2000-5000.



Hard drive PCB

We found 2 drives of the same make, one of which went kaput a few years ago. After a round of diagnostics, it was found out that we'd have to swap platters to extract the data from the spoilt disc.

The PCB here shows you the Micro-controller (MCU) and Flash BIOS chip. These in combination with information stored on the platter tell the MCU how to read/write data across the platter. So it is absolutely essential that you retain the Flash BIOS chip. Often swapping the platters along with the PCB does the trick and will allow you to read the data via the OS. In case your operating system asks you to initialize the HDD again, do not do so. Instead, boot up in a Linux distro and install ddrescue, it'll copy your data bit-by-bit onto another drive.

All these operations need to be followed in a clean room, which is a room with very few particulate matter and has continuous air flow. The air is filtered to such an extent that there are barely 12 particles(<0.1um)

within a cubic meter of air compared to 35 million particles which is present in a cubic meter or room air.

We remove the HDD covers, take out one half of the magnet and then the actuator limiter. We insert a small piece of rolled up paper in between the head armature to separate it from the platter surface, slowly slide the head away from the centre while rotating the platter counter-clockwise. Remove the metal ring holding the platters in place and transplant this platter into the donor casing. Do not touch either surface or bring any magnetic devices near it. Put back the metal ring and secure it. Slide in the head while rotating the platter counter-clockwise and remove the rolled up paper gently so that the head rests easily onto the resting area and doesn't slam onto the platter. The heads are very sensitive and proper care must be taken. Insert the limiter and the magnet. Secure the magnet and close the HDD casing. Flip the HDD and swap the PCB onto the donor drive. Connect the drive and check if your OS can access

CHECKING FOR PCB COMPATIBILITY

1. Same PCB type
2. Same PCB revision
3. If WD, match the DCM (Drive Configuration Matrix) till the 4-5th character. Interpreting the DCM -
 - a. Motor
 - b. Base
 - c. Latch
 - d. Bottom VCM
 - e. Media
 - f. Headstack
 - g. Actuator-Preamp
 - h. Top VCM
 - i. Separator

your data. You have a very narrow window to copy all your data over, specially if you performed the entire operation outside a clean room. If your OS cannot read the data but asks you to initialise the drive. Use ddrescue to perform a bit-by-bit copy. 

SIGNS OF A DYING HDD

1. Clicking sound
2. Frequent BSODs
3. S.M.A.R.T. Data
4. HDD Intermittently disappearing from the BIOS screen
5. Grinding noise
6. Abrupt OS hangs

problem, you may have to perform a live swap, which requires you to swap the PCB after connecting it and the OS has initialised the drive. This may work and if it does, keep an extra drive nearby to